

Sensor structure shapes the format of cognitive maps in navigation agents

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I. BACKGROUND AND MOTIVATION

A PointGoal navigation (PointNav) agent is a deep-RL system that, given a goal coordinate, must walk a 3D scene to reach it. Trained agents have been observed to develop *cognitive-map-like* internal representations of space inside their recurrent (LSTM) memory — consistent with the cognitive-map hypothesis [1] and documented even in agents *without* any visual input [2], with parallels to grid- and place-cell-like codes that have been reported in recurrent navigation models [3], [4]. The agent has two routes to the spatial information its policy needs to act each step: (i) recompute it from the current visual encoder output, or (ii) maintain it across time in the recurrent memory by integrating the GPS sensor signal. These routes compete for finite hidden-state capacity. The central question of this project is whether the structure of the visual input — its encoder bandwidth and spatial layout — influences which route the trained agent ends up using, and therefore the *format* the cognitive map takes.

Our working hypothesis (the *capacity-allocation* account) is that a bottleneck at the encoder leaves the integration route as the most viable mechanism, so the LSTM tends to carry a temporally-stable, linearly-readable position code in the policy-readable layer ($\mathbf{h}_2$). A rich encoder offers a per-step visual-feature route as an alternative; under policy-gradient training, capacity tends to flow toward whichever route the agent can exploit, and the linearly-readable component of $\mathbf{h}_2$ shrinks — though, as we show, the same hidden state still encodes position non-linearly. Same task, different cognitive-map format.

This framing is an evolution of the project’s original direction. The proposal and Report 1 focused on foveated vision specifically (testing whether the foveated condition shows compensatory memory). The post-retrain experiments below — particularly the foveated-logpolar control we added *after* Report 1 — pointed instead at encoder bandwidth in general as the substantive variable: foveation *per se* does not pick out a special regime. We adjusted the project scope and central question accordingly; §VI documents the shift in detail.

II. SETUP AND METHOD

We hold architecture, training, and the non-visual sensor stack constant across five conditions and vary only the visual input (Table I). All agents share a 3-layer LSTM (512-d) and the standard goal-in-start-frame / GPS / compass / close-to-goal sensor stack of [2]. We retrained all four sighted conditions from scratch under unified hyperparameters (250M frames each) to fix the inconsistency in our Report 1 runs. All achieve 96–99% task success and $\text{SPL} \in [0.85, 0.89]$ on

the same evaluation set, so any cross-condition representational difference is unlikely to be explained by a skill gap.

Table I
FIVE CONDITIONS, ORDERED BY ENCODER SPATIAL OUTPUT (SIZE OF THE VISUAL FEATURE MAP PER TIMESTEP). BLIND USES AN INDEPENDENTLY-RETRAINED CHECKPOINT AT 340M FRAMES.

Condition	Encoder out	Visual input
Blind	none	no RGB
Coarse	1×1	48×48 RGB downsampled
Foveated-logpolar	2×2	64×64 log-polar resampled
Foveated	4×4	256×256 Gaussian-blurred ($\sigma_{\max} = 8$)
Uniform	4×4	256×256 full-resolution RGB

The conditions span the encoder spatial-bandwidth axis: *blind* is the no-vision lower bound replicating [2]; *coarse* is a bottleneck condition with visual input but a 1×1 encoder that collapses spatial layout per timestep; *foveated-logpolar* is an intermediate-bandwidth control we added after Report 1 as a falsification test of the encoder-bandwidth account; *foveated* and *uniform* are the two rich-encoder endpoints, distinguished by whether the periphery is degraded.

For each condition we collect 500 deterministic-policy episodes and analyse the frozen top-layer hidden state $\mathbf{h}_2$ along three complementary methodological families. **Decodability** (*what does $\mathbf{h}_2$ encode?*) — linear (Ridge $\alpha=10$, what a linear policy head can read) and 2-layer MLP probes for GPS, lag- k probes for past position GPS_{t-k} , per-(unit, scene) Skaggs spatial information from the place-cell literature, and cross-training probes re-run at 5 checkpoints. **Intrinsic structure** (*do conditions share a code?*) — linear CKA [5] between conditions, with a Procrustes shape-distance panel queued. **Causal intervention** (*does the policy rely on $\mathbf{h}_2$?*) — cross-condition memory transplant (sighted-only currently; coarse / blind pairs deferred). All probes use 5-fold episode-level cross-validation. This report concentrates on the decodability column; intrinsic structure and causal intervention are advancing in parallel (§VII).

III. ENCODER BANDWIDTH SHAPES THE LINEAR AXIS

The cleanest summary of the post-retrain data is Figure 1 (solid line): linear GPS R^2 trends down smoothly as encoder spatial output grows. The trend gave us a falsification target: foveated-logpolar (a 2×2 -output rich-encoder control we added *after* Report 1) was predicted to land in the rich-encoder regime, and Figure 1 is consistent with that prediction — arguing against the alternative that foveation *specifically* (rather than encoder bandwidth in general) is what shapes the cognitive map. The other obvious alternative — that the linear probe simply lacks capacity in rich-encoder hidden states — is hard to reconcile with the fact that

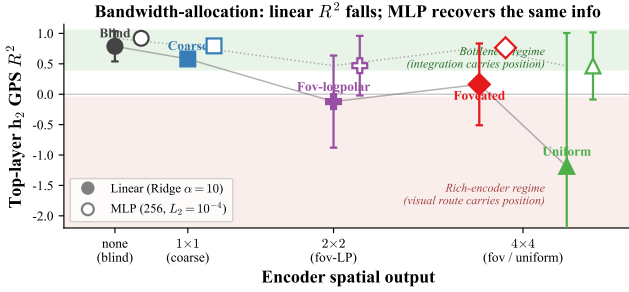


Figure 1. Top-layer GPS R^2 (mean $\pm \sigma$) vs. encoder spatial output. Solid: linear (Ridge) probe. Dashed: 2-layer MLP probe on the same hidden states. Blind point: preliminary 100-ep partial NPZ.

distance-to-goal, a same-dimensional control target, decodes at $R^2 > 0.8$ in every condition.

Position information appears not to be *destroyed* by a rich encoder. A non-linear MLP probe on the *same* hidden states recovers GPS in every condition (Figure 1, dashed), and the gap from linear to MLP widens with encoder bandwidth. The combined picture is consistent with the capacity-allocation account: the bottleneck (no encoder, or 1×1) leaves the integration route as the only viable mechanism, so the LSTM tends to store position in a linearly-readable axis; a rich encoder opens a per-step visual-feature route, capacity flows toward it, and the integrated code is no longer the linearly-readable axis — though, the MLP recovery suggests, position remains encoded. Format differs; total information shifts only modestly.

IV. ALLOCATION EMERGES ACROSS TRAINING

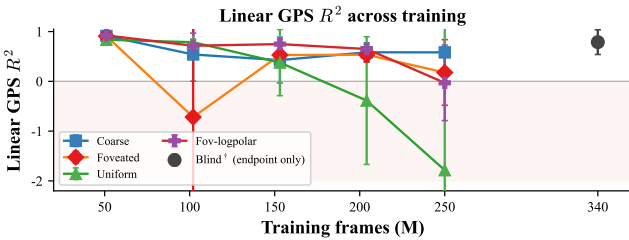


Figure 2. Linear GPS R^2 at 5 training checkpoints per sighted condition. The blind cross-training trajectory is deferred to the final report (intermediate blind checkpoints exist but have not yet been re-probed under the unified pipeline). Negative- R^2 outliers clipped to -2.0 .

The endpoint contrast leaves open whether bottleneck conditions ever even *learned* the visual route. Re-running the linear probe at intermediate checkpoints (Figure 2) suggests they did: all four sighted conditions begin training with a high linear top-layer GPS code, indicating that the linear axis is the *default* format the LSTM tends to acquire early on. What separates conditions is what happens *after*: bottleneck preserves the integrated code; the three rich-encoder conditions decay as the visual route consolidates, with decay rate ordered by encoder informativeness about position. To the extent the data support a mechanism, capacity allocation looks like a training-time process rather than an endpoint property; bottleneck conditions did not avoid the visual route

by accident, they kept the integration route on top because no usable visual alternative emerged.

V. BOTTLENECK INTEGRATES; RICH ENCODER REACTS

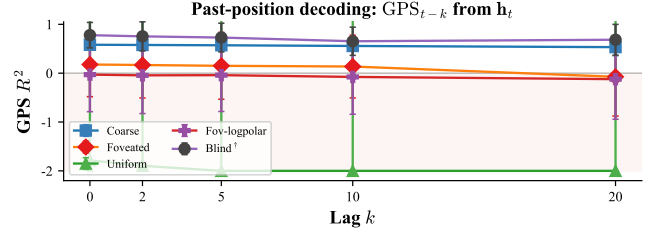


Figure 3. GPS R^2 for past-position probes GPS_{t-k} from current $\mathbf{h}_t$. Uniform’s mean R^2 falls below the visible range from $k=5$ onward (true values $-2.28, -3.01, -6.53$ at $k=5, 10, 20$); shown clipped at -2.0 . Blind point: preliminary 100-ep partial NPZ.

The same format difference shows up within an episode. Decoding past position from current $\mathbf{h}_t$ (Figure 3) shows bottleneck conditions sustain stable positive R^2 across the lag range while rich-encoder ones degrade rapidly: a smoothly-varying *integrated* code preserves past-position information; a *reactive* code keyed to current visual context does not. As a check against the obvious null — “maybe bottleneck conditions just have more place cells” — per-(unit, scene) Skaggs spatial information [2] is essentially condition-flat at $[1.16, 1.35]$ bits, well within the within-condition spread. The bandwidth-allocation signal we observe appears to live in the linear-readout axis rather than in the per-unit place-cell metric.

VI. DEVIATIONS FROM THE PROPOSAL

H1 reframed The proposal cast H1 as compensatory memory specifically in the foveated agent, and Report 1 (on partially-converged checkpoints with heterogeneous hyperparameters) reported foveated as the longest-memory condition. After unified retraining — which removed the apparent foveated specialness — and after adding foveated-logpolar as a falsification test, the data are more consistent with a cleaner encoder-bandwidth trend (Figure 1) under which foveated and uniform sit in the same rich-encoder regime. We now read the mechanism as encoder bandwidth rather than foveation specifically.

H3 de-scoped, methodology broadened H3 (epistemic gaze: a learned-gaze MLP targeting memory-weak regions) was de-scoped after the gaze decoder collapsed to a fixed point under pure navigation reward; a bounded- σ stochastic-gaze fix is in the codebase but not pursued this iteration. We extended the analytical methodology to import classical cognitive-neuroscience methods into the representational-analysis frame: Skaggs spatial information as a place-cell signature on $\mathbf{h}_2$, an SR-style predictive-horizon probe, Procrustes shape-metric and principal-angle subspace divergence, an excursion-forgetting protocol, and a leave-one-scene-out scene-invariance test. *The contribution shifts from a single epistemic-gaze deliverable toward applying classical cognitive-neuroscience metrics as a computational framework for representational analysis of deep-RL agents.*

VII. OPEN ISSUES AND ACTION ITEMS

Forward risks The long-form blind probe-collect is in flight; the 100-ep partial NPZ provides a fall-back if the full run does not converge in time. Single-seed per condition leaves per-condition seed-variance unbounded. The transplant matrix is incomplete without coarse / blind pairs (an RGB-shape architecture mismatch we resolved for the sighted-only pairs is the gating fix), so the test of the proposal’s H2 (cross-condition representational divergence) is deferred until that lands.

Bandwidth-allocation refinements (A1) complete the long-form blind probe-collect, populate the blind row across all lenses, and tighten Figures 1–3; (A2) retrain coarse with a 256×256 -padded input pipeline so the cross-condition memory-transplant matrix can include the bottleneck-vs-rich-encoder asymmetry test (closes the gap on H2 evidence); (A3) blind cross-training trajectory by re-probing intermediate blind checkpoints under the unified pipeline, completing Fig 2; (A4) two-seed validation on the coarse and uniform endpoints to bound seed-variance on the magnitude axis.

New cog-neuro analyses on existing h_2 (A5) allocentric occupancy decoder $f(\mathbf{h}_T, \mathbf{c}_T) \rightarrow 32 \times 32$ binary map — tests whether memory encodes a metric *map*, not just a position scalar (ground-truth occupancy already collected); (A6) MINE-based predictive-horizon $I(\mathbf{h}_t; \text{pos}_{t+k})$ for $k \in \{1, 5, 10, 20\}$, replacing decodability R^2 with nat-scale mutual information; (A7) successor-representation eigenbasis test linking h_2 PCs to $(\mathbf{I} - \gamma\mathbf{T})^{-1}$ eigenvectors; (A8) Sussillo–Barak fixed-point analysis to characterise format-divergence at the level of attractor structure; (A9) place-vs-view-cell dissociation, boundary-vector tuning, and ratemap drift; plus hierarchical scene-clustering as a relational cognitive-map test.

Scope-generalisation tests (A10) foveation strength sweep ($\sigma_{\max} \in \{2, 4, 12, 20\}$) probing the encoder–memory race as a *continuous* lever; (A11) transformer-based navigator under the same five conditions, probing attention / KV-cache for position — an architecture-agnostic falsification target; H3 reattempt with bounded- σ stochastic gaze if time permits.

VIII. IMPLICATIONS

If the capacity-allocation account holds up under the remaining tests, three implications follow. First, encoder bandwidth becomes a candidate *design lever* for shaping recurrent memory: deliberately restricting encoder spatial output during training may recruit integration-style memory that an unrestricted pipeline crowds out — a content-level rationale for foveation and bottleneck encoders, complementary to the usual compute-efficiency motivation [6]. Second, the encoder–memory race is in principle architecture-agnostic: vision–language models built atop rich foundation visual encoders may exhibit a similar crowd-out, plausibly contributing to the documented gap between fine-grained spatial reasoning and surface-level recognition in those

systems [7]. Third, the lens we apply — linear vs non-linear probes, lag- k , place-cell signatures — is portable: it provides a way to ask, of any recurrent agent, what *kind* of cognitive map (if any) it carries, with closer ties to comparative cognitive-neuroscience methodology than is standard in deep-RL representational analysis.

REFERENCES

- [1] E. C. Tolman, “Cognitive maps in rats and men,” *Psychological Review*, vol. 55, no. 4, pp. 189–208, 1948.
- [2] E. Wijmans, M. Savva, I. Essa, S. Lee, A. Morcos, and D. Batra, “Emergence of maps in the memories of blind navigation agents,” in *International Conference on Learning Representations (ICLR)*, 2023.
- [3] A. Banino, C. Barry, B. Uria, C. Blundell, T. Lillicrap, P. Mirowski, A. Pritzel, M. J. Chadwick, T. Degris, J. Modayil *et al.*, “Vector-based navigation using grid-like representations in artificial agents,” *Nature*, vol. 557, no. 7705, pp. 429–433, 2018.
- [4] C. J. Cueva and X.-X. Wei, “Emergence of grid-like representations by training recurrent neural networks to perform spatial localization,” in *International Conference on Learning Representations (ICLR)*, 2018.
- [5] S. Kornblith, M. Norouzi, H. Lee, and G. Hinton, “Similarity of neural network representations revisited,” in *International Conference on Machine Learning (ICML)*, 2019.
- [6] M. Pourrahimi and P. Bashivan, “Emergent brain-like representations in a goal-directed neural network model of visual search,” *bioRxiv preprint*, 2025.
- [7] S. K. Ramakrishnan, E. Wijmans, P. Krähenbühl, and V. Koltun, “Does spatial cognition emerge in frontier models?” in *International Conference on Learning Representations (ICLR)*, 2025.